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Chen

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(54) **NETWORK CONNECTOR**

(71) Applicant: **ELLAX CORPORATION**, Taoyuan (TW)

(72) Inventor: **Yu-Jun Chen**, Taoyuan (TW)

(73) Assignee: **ELLAX CORPORATION**, Taoyuan (TW)

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H01R 24/64 (2011.01)
H01R 107/00 (2006.01)

(52) **U.S. Cl.**
CPC ***H01R 12/58*** (2013.01); *H01R 24/64* (2013.01); *H01R 2107/00* (2013.01); *H01R 2201/04* (2013.01)

(58) **Field of Classification Search**
CPC H01R 12/58; H01R 12/716; H01R 12/724; H01R 24/62; H01R 24/64; H01R 2201/64; H01R 2107/00
See application file for complete search history.

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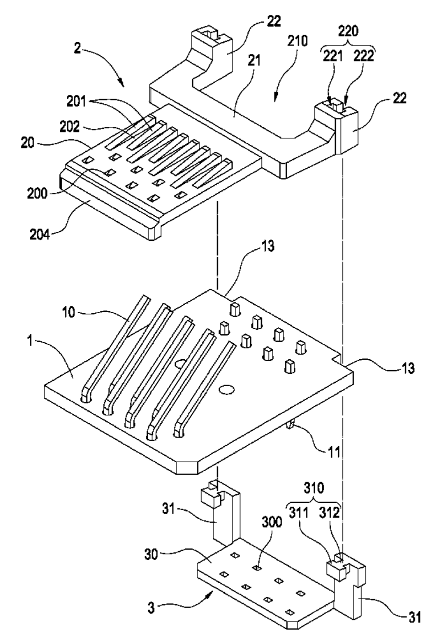
Primary Examiner — Gary F Paumen

(74) *Attorney, Agent, or Firm* — Chun-Ming Shih; HDLS IPR SERVICES

(57) **ABSTRACT**

A structural improvement of a network connector includes a circuit board, a terminal base and a pin base. The circuit board is disposed with terminals and pins. The terminal base has a terminal installing part and an assembling part. The terminal installing part is used for each terminal to pass. The assembling part has a hollow zone and two vertical slot seats. The pin base has a pin penetrating part corresponding to each pin, and two fastening parts. Each vertical slot seat has a vertical slot. Each fastening part has a slider corresponding to each vertical slot seat and used for fastening with the vertical slot, thus the pin base and the assembling part of the terminal base are fastened at a rear side of the circuit board.

10 Claims, 7 Drawing Sheets



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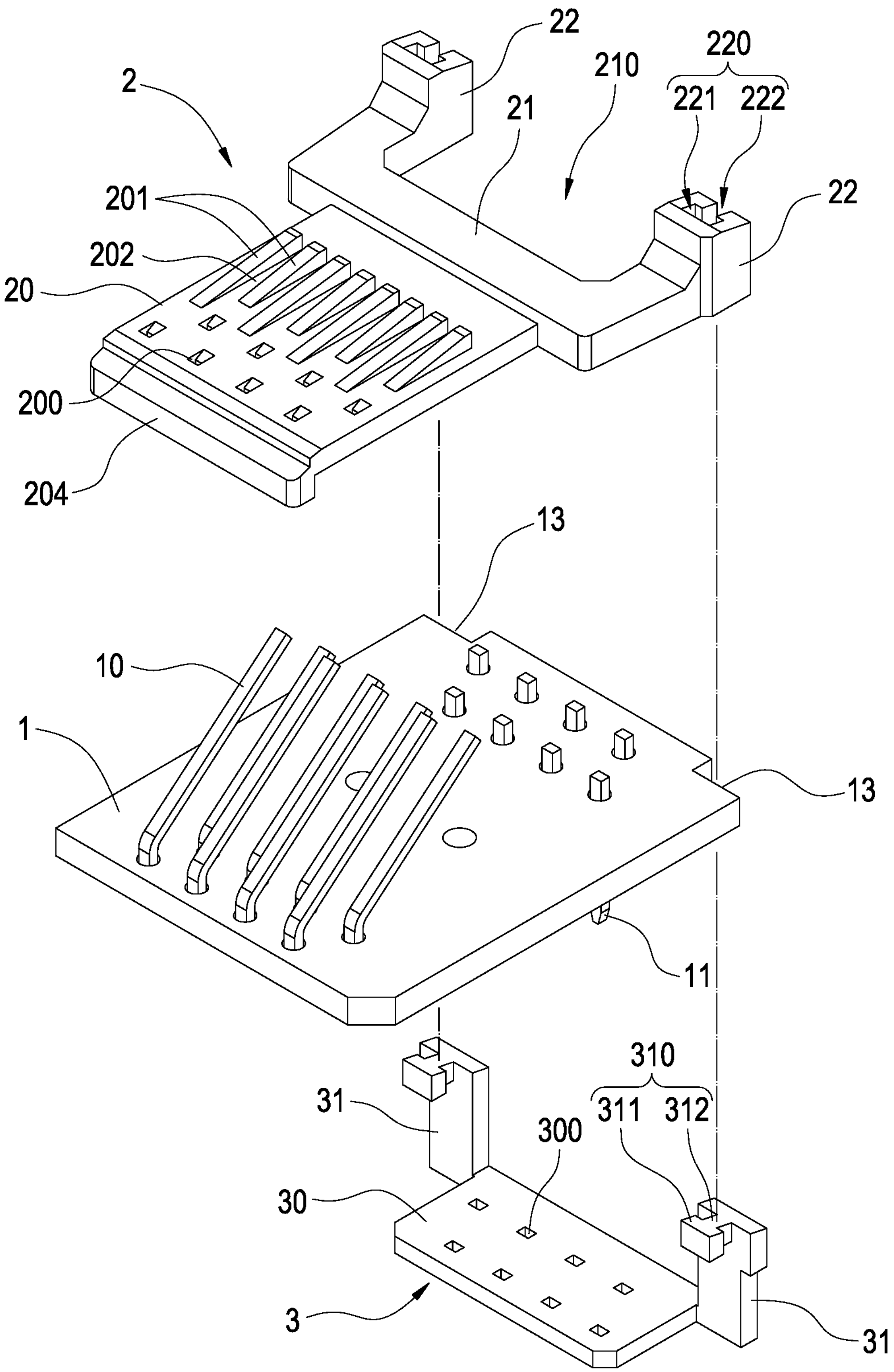
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FIG.1

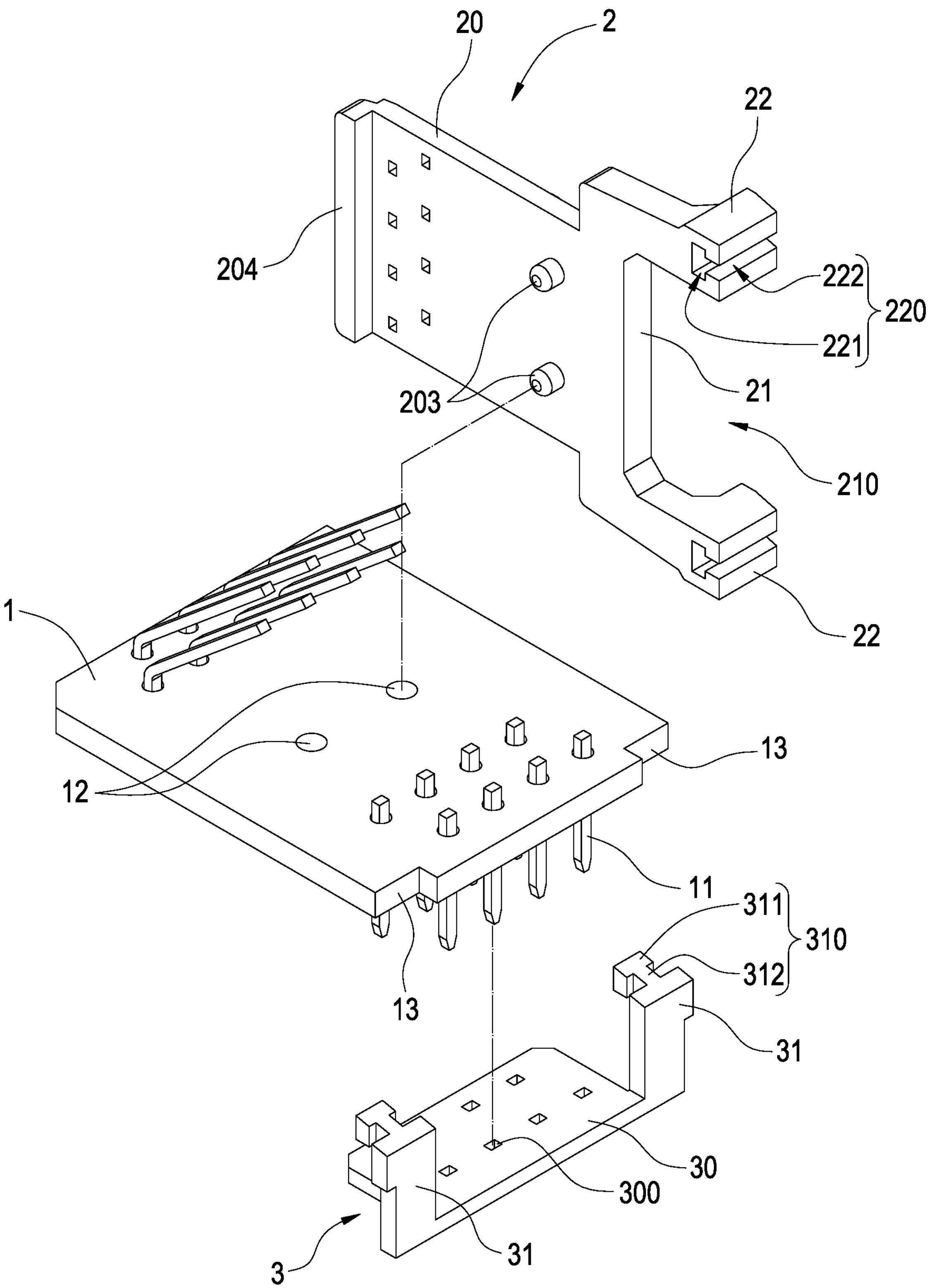
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FIG.2

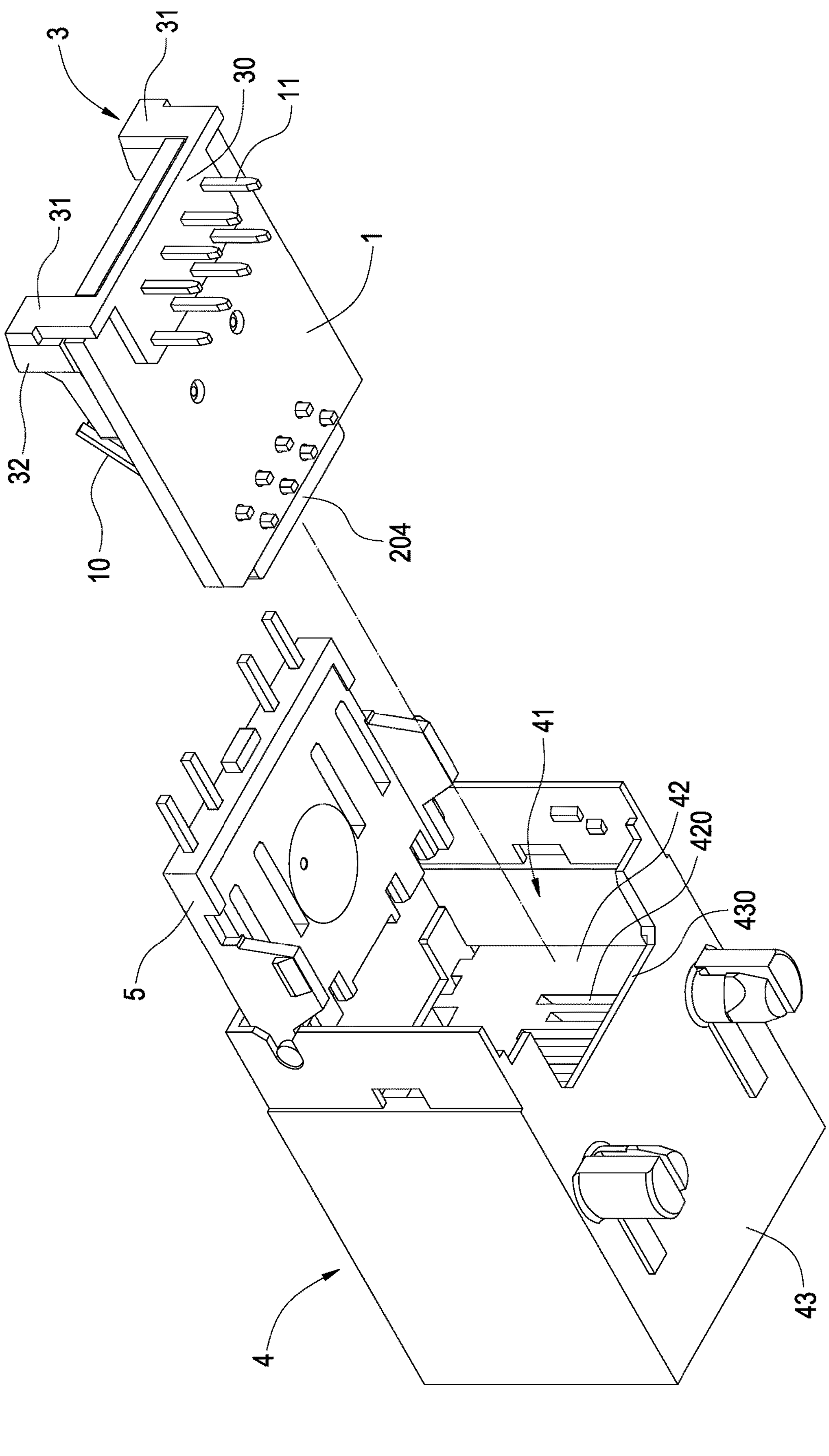

FIG.3

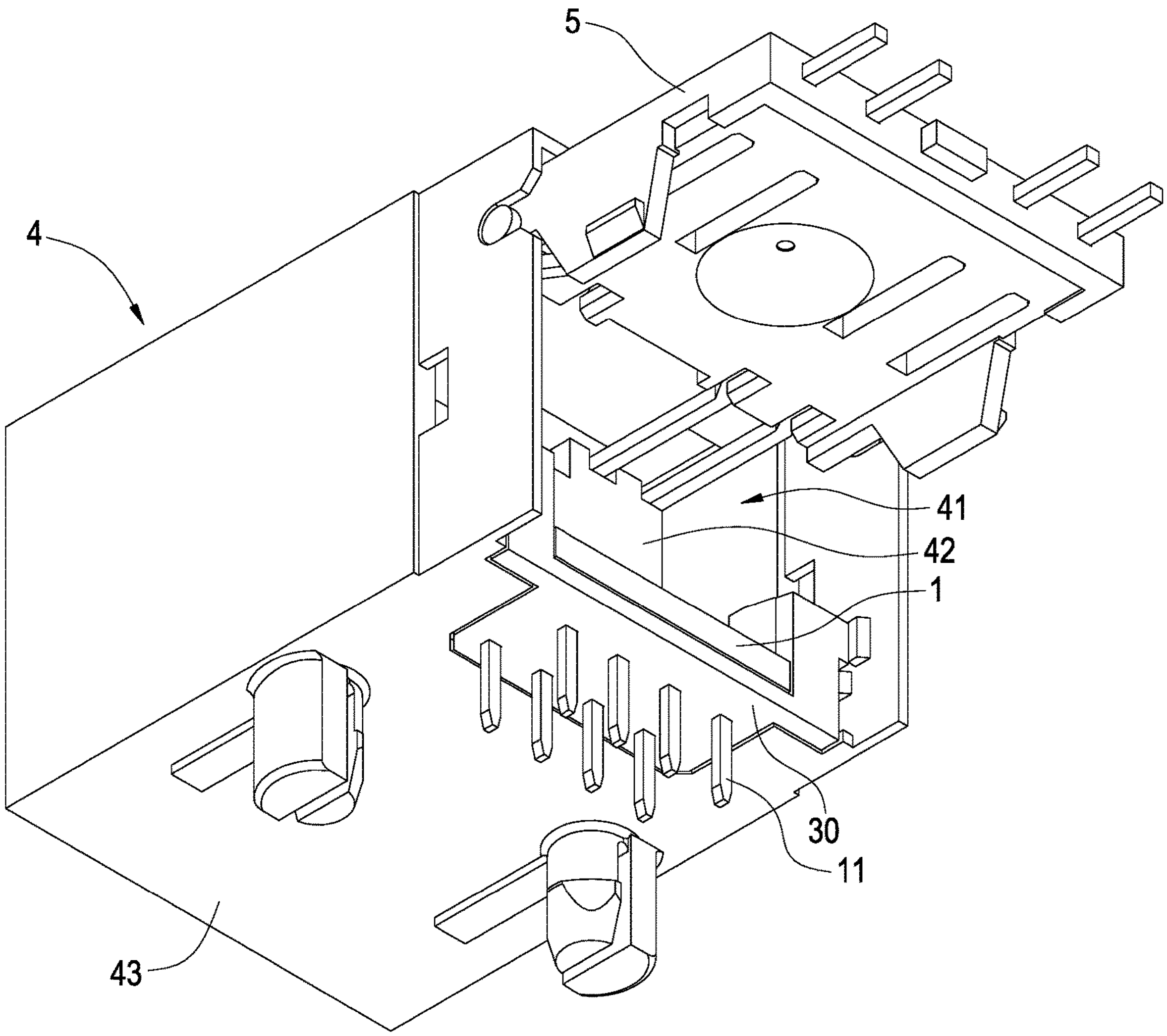

FIG.4

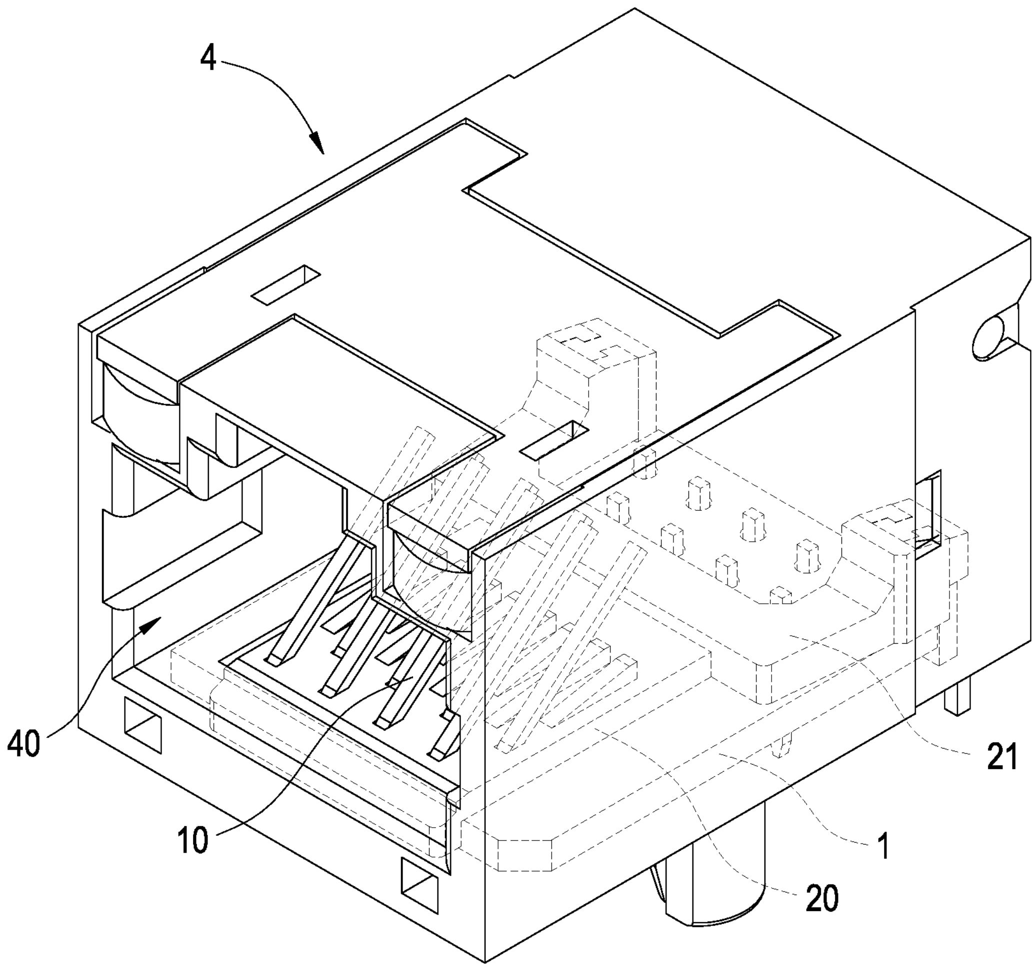

FIG.5

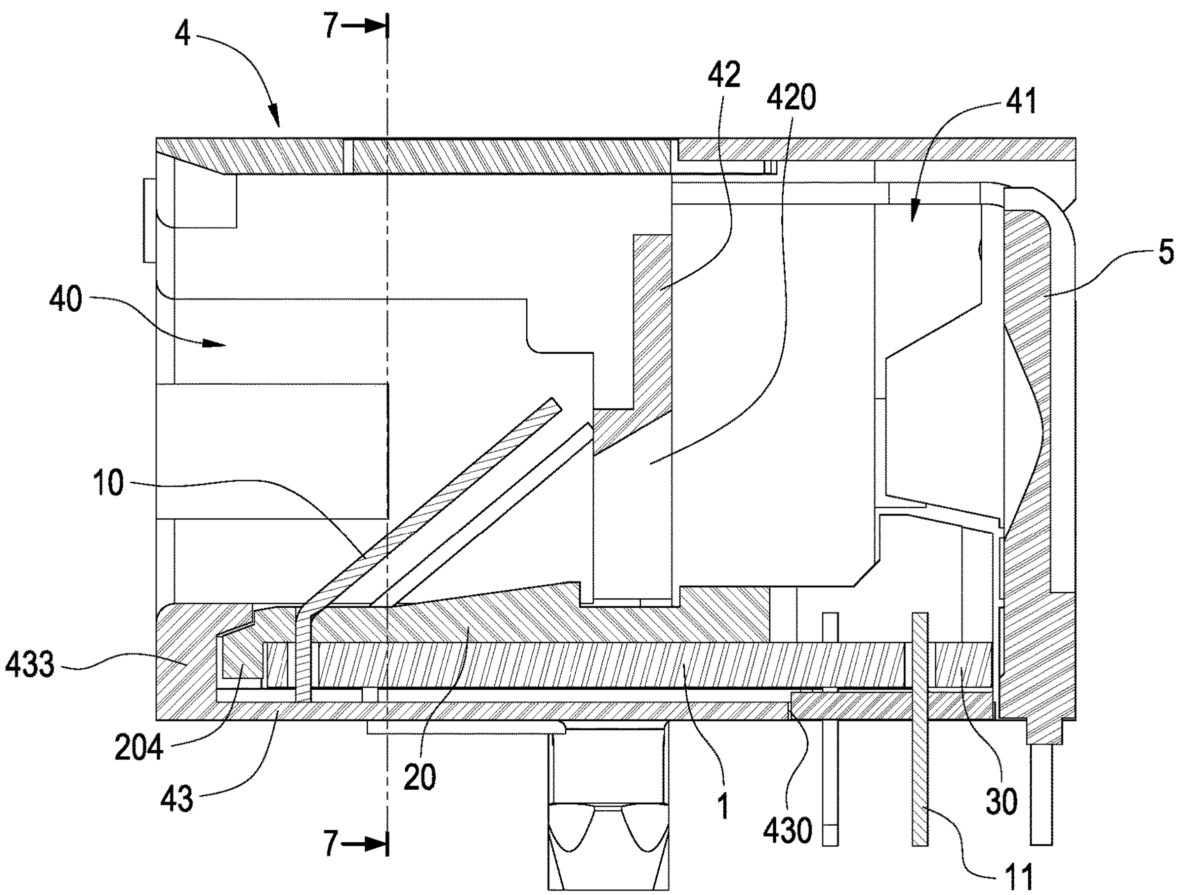

FIG.6

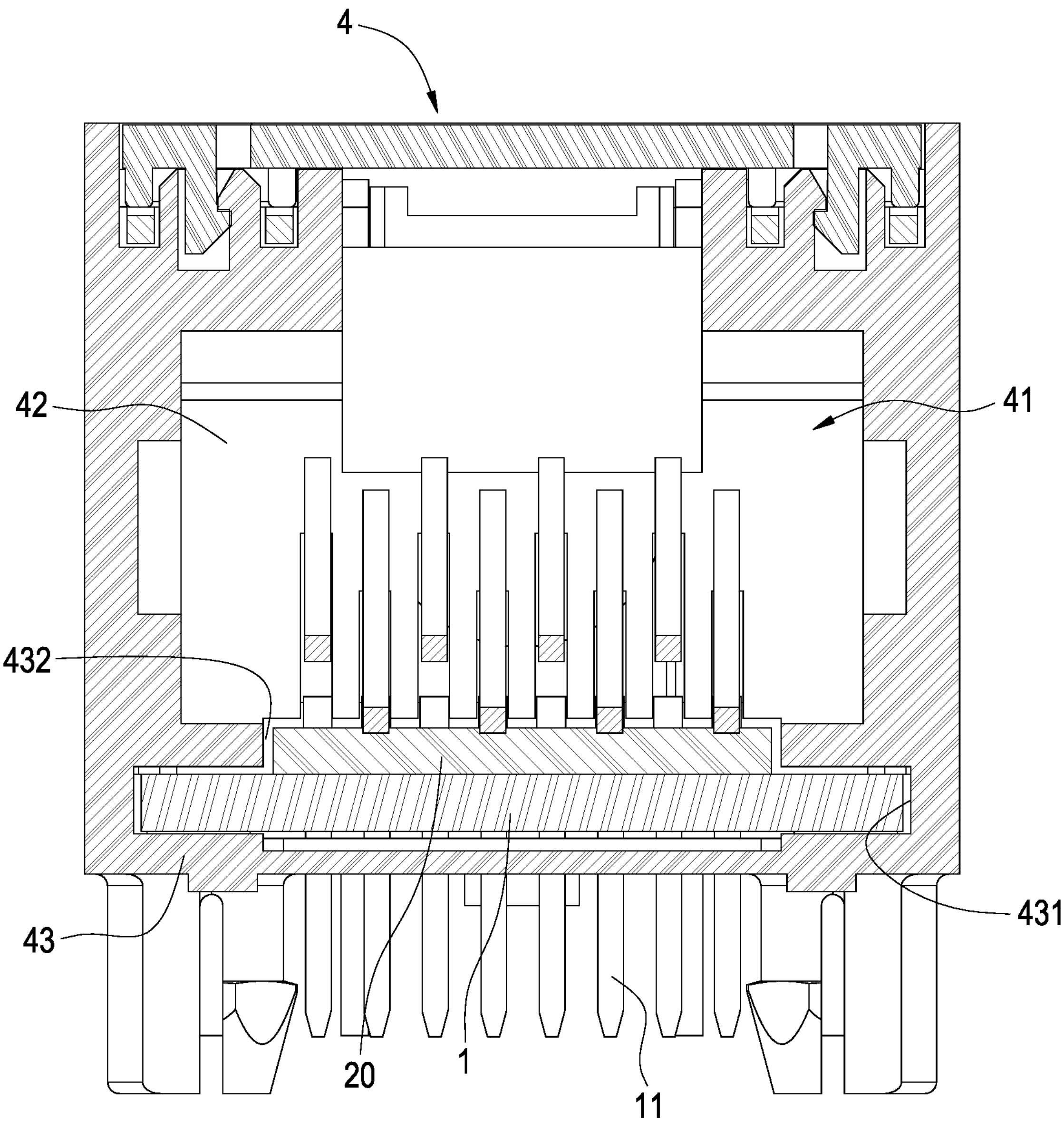

FIG.7

NETWORK CONNECTOR

BACKGROUND OF THE DISCLOSURE

Technical Field

The present disclosure relates to a connector, especially to a structural improvement of a network connector.

Description of Related Art

For avoiding being interfered by external electric waves during the signal transmission, a related-art network connector, such as a RJ45, is disposed with an electronic unit, such as a wave filter and a circuit thereof, for prevention. The wave filter is formed by soldering an electronic unit such as a coil on a circuit board, the disadvantage is that the processing procedure is complicated and occupies an unnecessary volume, thus the network connector may not be miniaturized. With the wave filtering unit being formed through an electronic unit such as an integrated circuit, the occupied volume is greatly reduced to achieve an objective to having a small volume.

However, when the wave filtering unit is soldered on the circuit board during the simplified procedure, terminals and pins required by the network connector are disposed on the circuit board. As such, the terminals and the pins themselves may lose the sufficient structural strength, and operations of inserting the network connector or the assembly soldering may be affected.

Accordingly, the applicant of the present disclosure has devoted himself for improving the mentioned shortages.

SUMMARY OF THE DISCLOSURE

The present disclosure is to provide a structural improvement of a network connector, in which terminals and pins of a network connector are integrated by utilizing a circuit board, and a terminal base is provided for reinforcing the assembling and structural strength, thus advantages of simplifying the structure and prolonging the service life are provided.

Accordingly, the present invention provides a structural improvement of a network connector, the network connector includes a circuit board, a terminal base disposed on a top surface of the circuit board and a pin base disposed on a bottom surface of the circuit board. The circuit board has a plurality of terminals protruded from the top surface thereof and a plurality of pins extended from the bottom surface thereof. The terminal base has a terminal installing part and an assembling part connected to a rear side of the terminal installing part. The terminal installing part is located above each of the terminals corresponding to the circuit board for each of the terminals to pass the terminal installing part. The assembling part has a hollow zone and two vertical slot seats, and the hollow zone is located on the top surface of the circuit board corresponding to each of the pins. The pin base has a pin penetrating part corresponding to each of the pins, and two fastening parts connected to a rear side of the pin penetrating part. The two fastening parts are upwardly extended from a rear side of the circuit board. Each vertical slot seat has a vertical slot. Each fastening part has a slider corresponding to each vertical slot seat and used for fastening the vertical slot, thus the pin base and the assembling part of the terminal base are fastened at the rear side of the circuit board.

BRIEF DESCRIPTION OF THE DRAWINGS

The features of the disclosure believed to be novel are set forth with particularity in the appended claims. The disclosure itself, however, may be best understood by reference to the following detailed description of the disclosure, which describes a number of exemplary embodiments of the disclosure, taken in conjunction with the accompanying drawings, in which:

FIG. **1** is a perspective exploded view according to the present disclosure;

FIG. **2** is another perspective exploded view according to the present disclosure;

FIG. **3** is a perspective exploded view showing the present disclosure being disposed in the housing;

FIG. **4** is a perspective view showing the assembly of the present disclosure being disposed in the housing;

FIG. **5** is another perspective view showing the assembly of the present disclosure being disposed in the housing;

FIG. **6** is a cross-sectional view showing the present disclosure being disposed in the housing; and

FIG. **7** is a cross-sectional view of FIG. **6** along a dashed line **7-7**.

DETAILED DESCRIPTION OF THE DISCLOSURE

The technical contents of this disclosure will become apparent with the detailed description of embodiments accompanied with the illustration of related drawings as follows. It is intended that the embodiments and drawings disclosed herein are to be considered illustrative rather than restrictive.

Please refer to FIG. **1** and FIG. **2**, FIG. **1** is a perspective exploded view according to the present disclosure and FIG. **2** is another perspective exploded view according to the present disclosure. The present disclosure provides a structural improvement of a network connector, the network connector includes a circuit board **1**, a terminal base **2** and a pin base **3**.

The circuit board **1** is disposed with a circuit or an electronic unit (not shown in figures) used for filtering waves. A plurality of terminals **10** are protruded from a top surface of the circuit board **1**, and a plurality of pins **11** are extended from a bottom surface of the circuit board **1**. In some embodiments, the terminals **10**, and the pins **11** are disposed at two sides of the circuit board **1**. For example, the terminals **10** are disposed at a front side of the circuit board **1**, and the pins **11** are disposed at a rear side of the circuit board **1**. Through an arranging relation within a certain distance, the desired circuit or the electronic unit used for filtering waves may be arranged on the circuit board **1** and relatively disposed between the terminals **10** and the pins **11**.

The terminal base **2** is disposed on the top surface of the circuit board **1** corresponding to each of the terminals **10** of the circuit board **1**. The terminal base **2** may be formed through a plastic ejection molding manner. The terminal base **2** includes a terminal installing part **20** and an assembling part **21** connected at a rear side of the terminal installing part **20**. The terminal installing part **20** is disposed with a positioning hole **200** corresponding to each of the terminals **10** and allowing each of the terminals **10** to pass, and the positioning hole **200** is correspondingly arranged above each of the terminals **10** relative to the circuit board **1**. Each of the terminals **10** on the top surface of the circuit board **1** is in an inclined status after passing the positioning hole **200**, thus each of the terminals **10** is pressed after being inserted and contacting with an engaging connector. As such, a wedge-shaped pad **201** is protruded from a rear side of each of the positioning holes **200** to prevent the terminal **10** from being overly pressed. A gap **202** is formed between the wedge-shaped pads **201** to dividing the wedge-shaped pads **201**. Moreover, the terminal base **2** is disposed with at least one convex piece **203** on a bottom surface of the terminal installing part **20** to be fastened in a fixing hole **12** of the circuit board **1**, and the fixing hole **12** is located between each of the terminals **10** and each of the pins **11**.

Details are provided as follows. The assembling part **21** of the terminal base **2** is disposed at a location of each of the pins **11** corresponding to the circuit board **1**. The assembling part **21** has a hollow zone **210** and two vertical slot seat **22** formed correspondingly. The hollow zone **210** is located on the top surface of the circuit board **1** corresponding to each of the pins **11** to avoid a soldered location of the pins **11** and the circuit board **1**. The two vertical slot seats **22** are disposed close to the rear side of the circuit board **1** and respectively formed with a vertical slot **220**, thus the pin base **3** is fastened with the assembling part **21** to make the pin base **3** and the assembling part **21** both be fastened at the rear side of the circuit board **1**.

Please refer to FIG. **3**, the pin base **3** is disposed on the bottom surface of the circuit board **1** corresponding to each of the pins **11** of the circuit board **1**. The pin base **3** may be formed through a plastic injection molding manner. The pin base **3** has a pin penetrating part **30** and two fastening parts **31** connected to a rear side of the pin penetrating part **30**. The pin penetrating part **30** is formed with a through hole **300** corresponding to each of the pins **11** and allowing each of the pins **11** to penetrate. As such, the stability of each of pins **11** being soldered on the circuit board **1** is increased via the pin penetrating part **30**. The two fastening parts **31** are upwardly extended from the rear side of the circuit board **1** to be corresponding to the two vertical slot seats **22** of the assembling part **21**, and the two fastening parts **31** respectively have a slider **310** fastened in the vertical slot **220**. In some embodiments, the slider **310** is formed in a convex status and has a wider bottom protrusion **311** and a narrower top protrusion **312**. The vertical slot **220** has a bottom recess **221** matched with the bottom protrusion **311**, and a top recess **222** matched with the top protrusion **312**. As such, with the geometrical design, the combining strength of the slider **310** and the vertical slot **220** is increased. In some embodiment, for example the slider **310** may be formed in a dovetail shape and the vertical slot **220** may be a matched dovetail-shaped slot.

For initially positioning the two fastening parts **31** and the rear side of the circuit board **1**, a rear edge of the circuit board **1** is concavely formed with notches **13** corresponding to the two fastening parts **31**.

Accordingly, the structural improvement of the network connector of the present disclosure is assembled through the aforesaid structure.

Please refer to FIG. **3**, FIG. **4**, FIG. **5** and FIG. **6**, the present disclosure further includes a housing **4**. An inserting port **40** is concavely disposed at a front side of the housing **4**, and the inserting port **40** allows the engaged connector to be inserted. A rear end of the housing **40** has an accommodating zone **41** defined concavely. A partition wall **42** located in the housing **4** is formed between the inserting port **40** and the accommodating zone **41**. The partition wall **42** is disposed with a plurality of bypass slots **420** corresponding to each of the terminals **10** to make the assembled circuit board **1**, the terminal base **2** and the pin base **3** be disposed from the accommodating zone **41** at the rear side of the housing **4**, thus each of the terminals **10** pass the bypass slot **420** to be located in the inserting port **40** of the housing **4**, as shown in FIG. **5** and FIG. **6**.

Please refer to FIG. **3** and FIG. **4**, a bottom part **43** of the housing **4** is formed with a slit **430** which is corresponding to an opening of the accommodating zone **41** and in a recessed status and allows the pin base **3** to be mounted, thus each of the pins **11** is directly exposed from the bottom part **43** of the housing **4**. Please refer to FIG. **6** and FIG. **7**, the bottom part **43** of the housing **4** is disposed with a first position limiting slot **431** allowing the circuit board **1** to be mounted and a second position limiting slot **432** exposed in the inserting port **40**. The second position limiting slot **432** allows the terminal installing part **20** to be mounted and positioned. The bottom part **43** of the housing **4** is further disposed with a pressing part **433** at a front side defined in the inserting port **40**, and a front edge of the terminal installing part **20** is protruded with a pressing edge **204** which is downwardly pressed by the pressing part **433**, thus the terminal base **2** is prevented from tilting, and the inserting and contacting operation of the terminals **10** is not affected.

While this disclosure has been described by means of specific embodiments, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope and spirit of this disclosure set forth in the claims.

What is claimed is:

1. A network connector, comprising:

a circuit board, comprising a plurality of terminals disposed protrusively from a top surface thereof and a plurality of pins extended from a bottom surface thereof;

a terminal base, disposed on the top surface of the circuit board and comprising a terminal installing part and an assembling part connected to a rear side of the terminal installing part, wherein the terminal installing part is located above the terminals corresponding to the circuit board for the terminals to pass therethrough, the assembling part comprises two vertical slot seats and a hollow zone defined between the vertical slot seats, and the hollow zone is defined corresponding to the pins and located on the top surface of the circuit board; and

a pin base, disposed on the bottom surface of the circuit board corresponding to the pins and comprising a pin penetrating part and two fastening parts connected to a rear side of the pin penetrating part, wherein the two fastening parts are upwardly extended from a rear side of the circuit board;

wherein, each vertical slot seat comprises a vertical slot, each fastening part comprises a slider fastened to the vertical slot corresponding to each vertical slot seat, and the circuit board is sandwiched between the pin base and the assembling part of the terminal base at the rear side thereof.

2. The network connector according to claim **1**, wherein the circuit board comprises at least one fixing hole, and the terminal installing part comprises at least one convex piece disposed on a bottom surface thereof and fastened in the fixing hole.

3. The network connector according to claim **1**, wherein the circuit board comprises multiple notches concavely defined on a rear edge thereof corresponding to the two fastening parts.

4. The network connector according to claim **1**, wherein the terminal installing part comprises multiple positioning holes for the terminals to pass through, and the terminals of the circuit board pass through the positioning holes to be in an inclined status.

5. The network connector according to claim **4**, wherein the terminal installing part comprises a wedge-shaped pad disposed protrusively on a rear side of each of the positioning holes.

6. The network connector according to claim **1**, wherein the pin penetrating part comprises multiple through holes for the pins to penetrate.

7. The network connector according to claim **1**, further comprising a housing, wherein the housing comprises an inserting port defined concavely on a front side thereof and an accommodating zone defined concavely on a rear end thereof, the circuit board, the terminal base and the pin base are disposed in the housing from the accommodating zone, and the terminals are located in the inserting port of the housing.

8. The network connector according to claim **7**, wherein the housing comprises a partition wall located between the inserting port and the accommodating zone, and the partition wall comprises a plurality of bypass slots corresponding to the terminals.

9. The network connector according to claim **7**, wherein the housing comprises a slit concavely defined on a bottom part thereof corresponding to an opening of the accommodating zone for the pin base to be mounted, and the pins are exposed from the bottom part of the housing.

10. The network connector according to claim **7**, wherein the housing comprises a first position limiting slot defined on a bottom part thereof for the circuit board to be mounted and a second position limiting slot defined in the inserting port for the terminal installing part to be mounted, the housing further comprises a pressing part disposed on the bottom part thereof at a front side in the inserting port, and the terminal installing part comprises a pressing edge disposed protrusively on a front edge thereof and downwardly pressed by the pressing part.

* * * * *